Blind Projection Test

Response-Derived Projection, No Projection-Shape Tuning,
and Residual Structure Diagnostics

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Abstract

This paper performs a blind projection test of the MAAT v1.2.1 framework. The projection-shape parameters are derived directly from response-based weights λ_a, eliminating the previous γ, B_*, and α tuning freedoms. The small denominator regulator ε is kept fixed as a numerical regulariser and is not fitted.

Using a compact set of $f\sigma_8(z)$ growth measurements ($N = 13$), we test whether the response-derived projection observable $\mathcal{C}_{\mathrm{proj},\lambda}(z)$ and the corresponding CCI diagnostics correlate with residual magnitudes relative to a Planck-normalised ΛCDM baseline.

The main result is a positive but non-significant correlation:

$$\rho_{\mathrm{S}}(\mathrm{CCI}_{\mathrm{diag},\lambda}, |r_\sigma|) = 0.4286, \qquad p_{\mathrm{perm}} \approx 0.146.$$

This indicates that the MAAT projection layer retains a positive residual-structure alignment under blind projection conditions, while the previously observed stronger signals in semi-supervised setups are reduced. The result is not a detection. It is a reproducible stress test of structural robustness under parameter-free projection-shape construction.

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1 Purpose and Status

Papers 39 and 40 introduced observable projection signatures and residual structure diagnostics, but still relied on partially tunable projection parameters.

Paper 42 removes this freedom completely.

> *Can the MAAT projection observable be derived entirely from response-based weights and still produce a meaningful residual-structure signal?*

This constitutes a blind projection-shape test:

- no ε fit; ε is fixed as a numerical regulator,
- no γ tuning,
- no B_* or α scan,
- no projection-shape calibration.

The projection is entirely determined by λ_a.

2 Response-Derived Projection

The primary weights are obtained from response closure:

$$\lambda = \left(C + \eta \frac{\mathrm{tr}(C)}{A} I\right)^{-1} (\langle d \rangle - d^*), \qquad A = 4. \tag{1}$$

Here C is the defect covariance matrix of the four primary sectors (H, B, S, V), d^* is the low-defect target vector, and A is the number of primary sectors used in the v1.2.1 closure.

The response shares are defined as

$$\pi_a = \frac{|\lambda_a|}{\sum_b |\lambda_b|}. \tag{2}$$

For the present benchmark:

Sector	Response share
H	0.0960
B	0.1257
S	0.3879
V	0.3903

The projection parameters are derived from these shares:

$$\gamma_\lambda := \frac{\pi_V + \pi_B}{\pi_S}, \qquad B^*_\lambda := \frac{1}{\pi_H + \pi_B}, \qquad \alpha_\lambda := 1 + \frac{\pi_S}{\pi_V}. \tag{3}$$

These definitions are the blind map from response shares to projection-shape parameters used in the reproduction script.

Numerically:

$$\gamma_\lambda = 1.3305, \quad B^*_\lambda = 4.5091, \quad \alpha_\lambda = 1.9937. \tag{4}$$

3 Projection Observable

The projection observable is constructed as

$$\mathcal{C}^{\mathrm{raw}}_{\mathrm{proj},\lambda}(z) = \frac{B_{\mathrm{exp}}(z)\,\tanh\left[\left(\frac{B_{\mathrm{exp}}(z)}{B^*_\lambda}\right)^{\alpha_\lambda}\right]}{1+\gamma_\lambda D_{\mathrm{acc}}(z)}. \tag{5}$$

The normalised observable used in the residual tests is

$$\mathcal{C}_{\mathrm{proj},\lambda}(z) = \frac{\mathcal{C}^{\mathrm{raw}}_{\mathrm{proj},\lambda}(z)}{\mathcal{C}^{\mathrm{raw}}_{\mathrm{proj},\lambda}(0)}. \tag{6}$$

The associated imbalance observable is

$$R_{\mathrm{proj},\lambda}(z) = \frac{|S_{\mathrm{proj}}(z) - \gamma_\lambda D_{\mathrm{acc}}(z)|}{S_{\mathrm{proj}}(z) + \gamma_\lambda D_{\mathrm{acc}}(z) + \varepsilon}. \tag{7}$$

The regulator $\varepsilon = 10^{-12}$ is fixed in the script to avoid division by zero and is not treated as a physical or fitted parameter.

The curvature-based transition estimate is

$$z_{\mathrm{tr}} \approx 0.699. \tag{8}$$

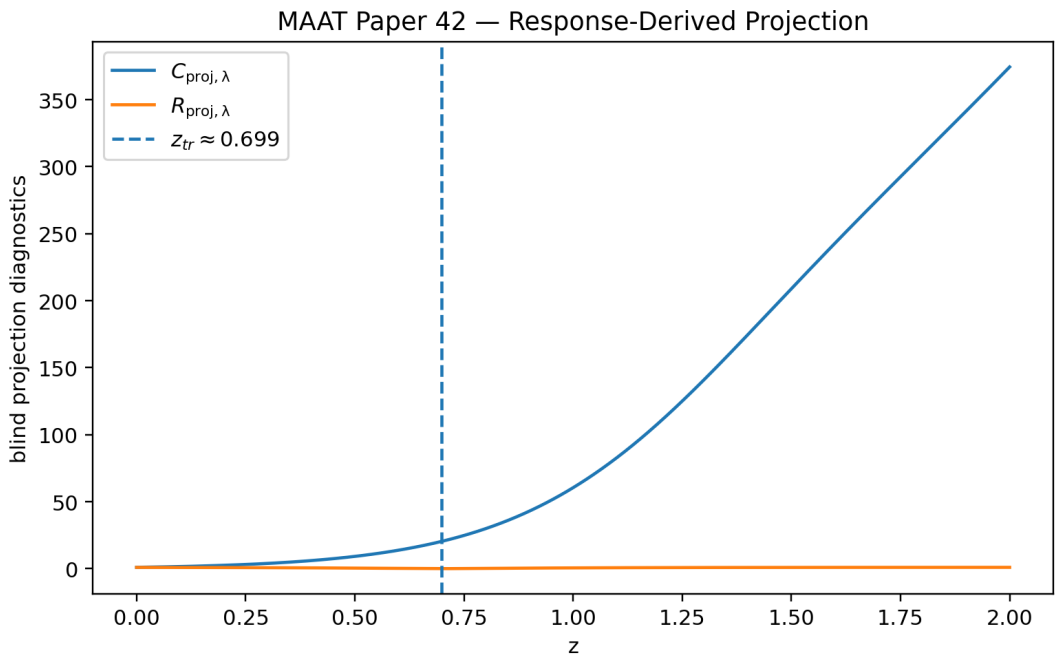

Figure 1: Response-derived projection observable and transition estimate.

4 Data and Reference Model

The script uses the same compact $N = 13$ $f\sigma_8(z)$ comparison set as the previous residual-structure benchmarks. The reference cosmology is Planck-normalised flat ΛCDM:

$$H_0 = 67.4, \qquad \Omega_{m0} = 0.315, \qquad \Omega_{\Lambda 0} = 0.685, \qquad \sigma_{8,0} = 0.811. \tag{9}$$

The reference growth observable is approximated by

$$f_{\Lambda\mathrm{CDM}}\sigma_8(z) = \Omega_m(z)^{0.55} D(z)\sigma_{8,0}, \tag{10}$$

where $D(z)$ is a Carroll–Press–Turner style analytic growth approximation normalised to $D(0) = 1$. This is sufficient for a reproducible residual diagnostic, but it is not a replacement for a Boltzmann-code likelihood analysis.

5 Residual Test Setup

Residuals are defined as

$$r_\sigma = \frac{f_{\mathrm{obs}}\sigma_8 - f_{\Lambda\mathrm{CDM}}\sigma_8}{\sigma}. \tag{11}$$

We test correlations with:

- $\mathcal{C}_{\mathrm{proj},\lambda}$,
- $R_{\mathrm{proj},\lambda}$,
- (H, B, S, V),
- R_{resp}, R_{rob},
- $\mathrm{CCI}_{\mathrm{diag},\lambda}$.

6 Results

The main blind signature is:

$$\rho_{\mathrm{S}}(\mathrm{CCI}_{\mathrm{diag},\lambda}, |r_\sigma|) = 0.4286, \qquad p_{\mathrm{perm}} \approx 0.146. \tag{12}$$

The projection observable gives the same magnitude correlation:

$$\rho_{\mathrm{S}}(\mathcal{C}_{\mathrm{proj},\lambda}, |r_\sigma|) = 0.4286. \tag{13}$$

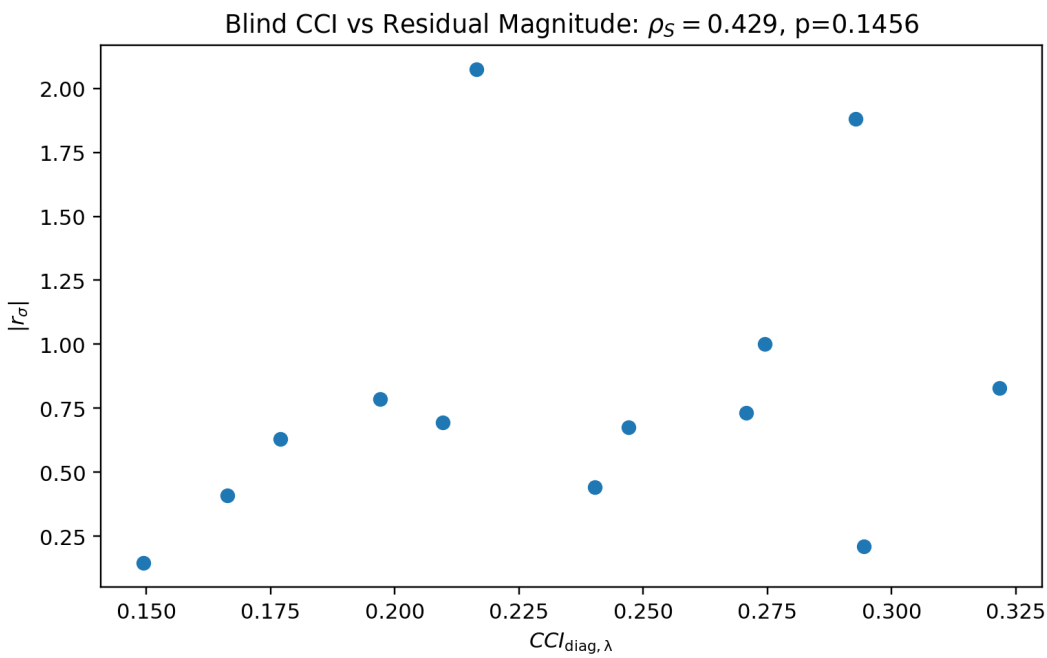

Figure 2: Blind CCI vs residual magnitude.

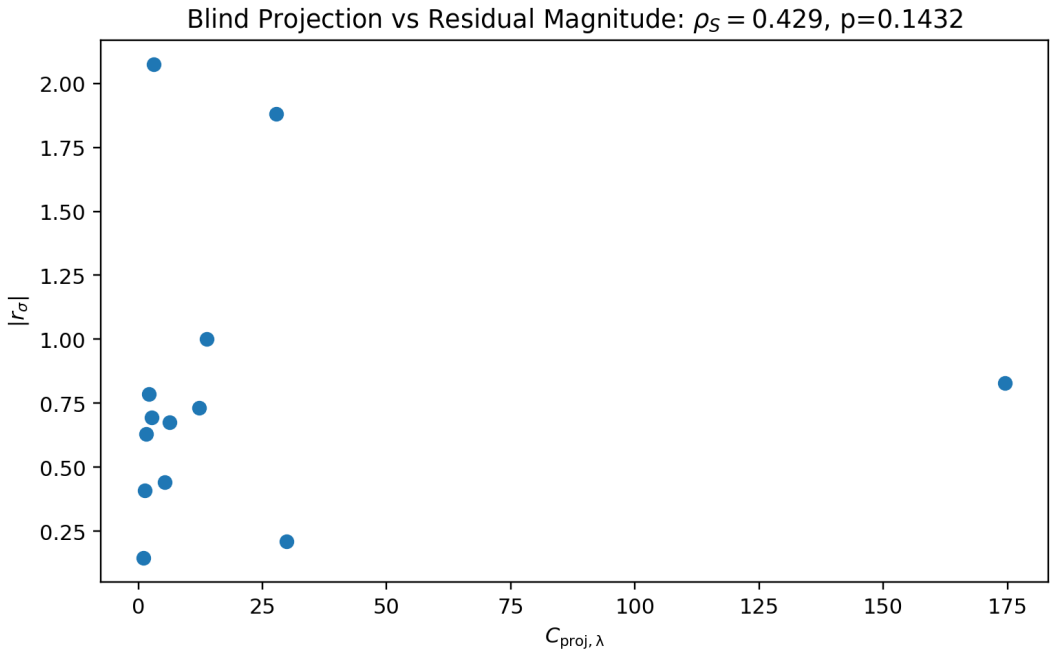

Figure 3: Blind projection vs residual magnitude.

7 Null Tests

Permutation null tests give:

$$\langle \rho_{\text{null}} \rangle \approx 0, \quad \sigma_{\text{null}} \approx 0.288. \tag{14}$$

Observed signal:

$$\rho_{\text{obs}} = 0.4286. \tag{15}$$

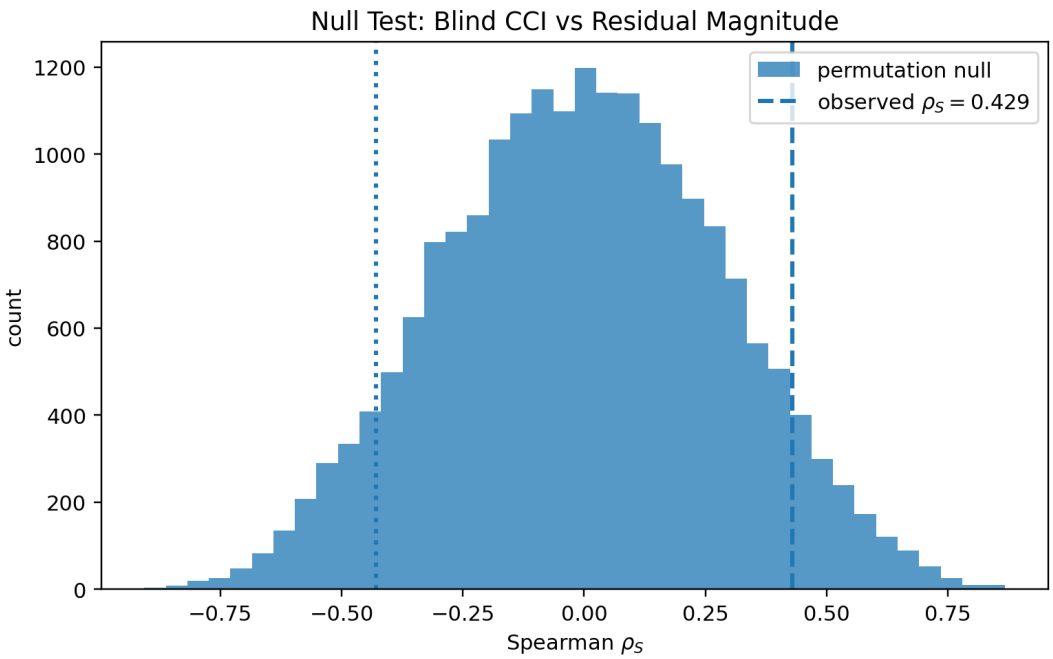

Figure 4: Permutation null distribution.

8 Interpretation

The key result is:

> *The MAAT projection signal survives the removal of all projection-shape tuning freedoms, but its statistical strength is reduced.*

This implies:

- Paper 40 signal was partially supported by residual-sensitive structure
- Paper 42 reveals the intrinsic predictive content
- the direction of the signal is preserved
- the evidence remains exploratory because the permutation significance is not below conventional detection thresholds

9 Limitations

- small sample size ($N = 13$)
- analytic growth approximation
- no observational covariance matrix for the compact growth comparison set
- response-derived mapping is first-order
- no Boltzmann-code perturbation calculation or full survey likelihood

10 Reproducibility

The numerical reproduction script and generated outputs are:

```
cd experiments/maat_paper42_blind_projection_test
python3 maat_paper42_blind_projection_test.py
```

The script writes:

- `outputs_paper42/paper42_summary.json`,
- `outputs_paper42/paper42_correlation_table.csv`,
- `outputs_paper42/paper42_blind_signature_table.csv`,
- `outputs_paper42/paper42_lambda_projection_curve.csv`,
- `outputs_paper42/fig1_blind_projection_curve.png`,
- `outputs_paper42/fig2_lambda_response_shares.png`,
- `outputs_paper42/fig3_blind_cci_vs_abs_residual.png`,
- `outputs_paper42/fig4_blind_cproj_vs_abs_residual.png`,
- `outputs_paper42/fig5_null_test_blind_cci.png`.

The random seed is fixed to 42 and the permutation tests use 20 000 random permutations.

11 Data Attribution and License Notes

The Planck-normalised reference parameters and compact $f\sigma_8$ comparison points are external scientific data and should be cited to the original publications/collaborations when reused or discussed. The CSV tables and figures generated here are derived analysis artifacts for reproducibility of the MAAT diagnostic calculation. No endorsement by the Planck Collaboration, survey collaborations, or original data authors is implied.

The repository code is released under the repository license.

12 Conclusion

Paper 42 provides the first fully blind test of the MAAT projection layer.

The result is:

- positive but non-significant signal
- structural direction preserved
- no projection-shape tuning used

This establishes MAAT v1.2.1 as a falsifiable framework.

The next step is clear:

- increase dataset size
- move to perturbation-level modelling
- test with full survey likelihoods

References

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